Online System-Level Synthesis for Prediction-Adaptive Networked Control (Supplementary Material)

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NOTATION AND PAPER REVIEW

Let $[N] = \{1, \dots, N\}$ and $T \geq d \geq 1$. The state and input dimensions are n_x and n_u. For $Q \succ 0$, $R \succ 0$, and $P \succeq 0$, define

$$J^\star(\mathbf{w}) := \min_{x,u} \sum_{t=0}^{T-1} (x_t^\top Q x_t + u_t^\top R u_t) + x_T^\top P x_T, \quad \text{(S.1a)}$$

$$\text{s.t. } x_{t+1} = Ax_t + Bu_t + w_t, \quad x_0 = \bar{x}. \quad \text{(S.1b)}$$

The lifted vectors are

$$\mathbf{x} = \mathrm{col}(x_0, \dots, x_T), \qquad \mathbf{u} = \mathrm{col}(u_0, \dots, u_{T-1}),$$
$$\mathbf{w} = \mathrm{col}(x_0, w_0, \dots, w_{T-1}), \quad \widehat{\mathbf{w}} = \mathrm{col}(0, \widehat{w}_0, \dots, \widehat{w}_{T-1}).$$

Thus $[\mathbf{w}]_0 = x_0$, $[\widehat{\mathbf{w}}]_0 = 0$, and $([\mathbf{w}]_k, [\widehat{\mathbf{w}}]_k) = (w_{k-1}, \widehat{w}_{k-1})$ for $1 \leq k \leq T$. Set $\mathbf{Q} = \mathrm{diag}(Q, \dots, Q, P), \mathbf{R} = I_T \otimes R, \mathbf{W} = \mathrm{diag}(\mathbf{Q}^{1/2}, \mathbf{R}^{1/2})$. For a feasible policy π, let $J_T(\pi, \mathbf{w}; \widehat{\mathbf{w}})$ be the objective in (S.1a) evaluated on its executed trajectory, and define

$$\mathtt{Reg}_T(\pi, \mathbf{w}; \widehat{\mathbf{w}}) = J_T(\pi, \mathbf{w}; \widehat{\mathbf{w}}) - J^\star(\mathbf{w}). \quad \text{(S.2)}$$

Let Z be the lifted block downshift, and set

$$\mathbf{A} = I_{T+1} \otimes A, \qquad \mathbf{B} = \begin{bmatrix} I_T \otimes B \\ 0 \end{bmatrix}.$$

The fixed CLM representation and its achievability equations are

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{u} \end{bmatrix} = \mathbf{\Phi}\mathbf{w} + \widehat{\mathbf{\Phi}}\widehat{\mathbf{w}}, \qquad \mathbf{\Phi} = \begin{bmatrix} \mathbf{\Phi}^x \\ \mathbf{\Phi}^u \end{bmatrix}, \quad \widehat{\mathbf{\Phi}} = \begin{bmatrix} \widehat{\mathbf{\Phi}}^x \\ \widehat{\mathbf{\Phi}}^u \end{bmatrix}, \quad \text{(S.3)}$$

$$\begin{bmatrix} I - Z\mathbf{A} & -Z\mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{\Phi} & \widehat{\mathbf{\Phi}} \end{bmatrix} = \begin{bmatrix} I & 0 \end{bmatrix}. \quad \text{(S.4)}$$

The causal response $\mathbf{\Phi}^{\mathrm{off}}$ is held fixed. For $0 \leq \ell \leq T$, let

$$\underline{\ell} = \max\{0, \ell - d\}, \quad \bar{\ell} = \min\{T, \ell + d - 1\}, \quad \mathcal{I}_\ell = \{\underline{\ell}, \dots, \bar{\ell}\}.$$

The linear space $\mathcal{F}_d$ prescribes $\widehat{\Phi}^a_{\ell,k} = 0$ for $k \notin \mathcal{I}_\ell$. The optimization results below use this support constraint. Achievability (S.4) is a separate restriction. Every sum over $a \in \{x, u\}$ omits the input term when $\ell = T$. For the fixed reference $\widehat{\mathbf{\Phi}}^{\mathrm{off}} \in \mathcal{F}_d$, define

$$r^a_\ell(\widehat{\mathbf{\Phi}}, \widehat{\mathbf{w}}) := \sum_{k \in \mathcal{I}_\ell} \left(\widehat{\Phi}^a_{\ell,k} [\widehat{\mathbf{w}}]_k - \widehat{\Phi}^{a,\mathrm{off}}_{\ell,k} [\mathbf{w}]_k \right), \quad \text{(S.5)}$$

$$\widetilde{h}_s(\widehat{\mathbf{\Phi}}) := \frac{1}{2} \sum_{\ell=0}^{s} \sum_{a \in \{x,u\}} \|r^a_\ell(\widehat{\mathbf{\Phi}}, \widehat{\mathbf{w}})\|_2^2 + \frac{\lambda}{2} \sum_{\ell=0}^{s} \sum_{k \in \mathcal{I}_\ell} \sum_{a \in \{x,u\}} \|\widehat{\Phi}^a_{\ell,k} - \widehat{\Phi}^{a,\mathrm{off}}_{\ell,k}\|_F^2, \quad \lambda > 0. \quad \text{(S.6)}$$

The prediction and disturbance records are fixed in these losses. The prefix operator and hindsight minimizer are

$$(\mathsf{P}_s \widehat{\mathbf{\Phi}})^a_{\ell,k} := \begin{cases} \widehat{\Phi}^a_{\ell,k}, & \ell \leq s, \\ 0, & \ell > s, \end{cases} \quad \text{(S.7)}$$

$$\widehat{\mathbf{\Phi}}^\diamond := \arg\min_{\widehat{\mathbf{\Phi}} \in \mathcal{F}_d} \widetilde{h}_T(\widehat{\mathbf{\Phi}}). \quad \text{(S.8)}$$

For $\ell \leq s$ and $k \in \mathcal{I}_\ell$, differentiation gives

$$\nabla_{\widehat{\Phi}^a_{\ell,k}} \widetilde{h}_s(\widehat{\mathbf{\Phi}}) = r^a_\ell(\widehat{\mathbf{\Phi}}, \widehat{\mathbf{w}}) [\widehat{\mathbf{w}}]_k^\top + \lambda(\widehat{\Phi}^a_{\ell,k} - \widehat{\Phi}^{a,\mathrm{off}}_{\ell,k}). \quad \text{(S.9)}$$

Initialize $\widehat{\mathbf{\Phi}}(t) = \widehat{\mathbf{\Phi}}^{\mathrm{off}}$ for $0 \leq t \leq d$, and update

$$\widehat{\mathbf{\Phi}}(t+1) = \Pi_{\mathcal{F}_d} \left[\widehat{\mathbf{\Phi}}(t) - \eta \nabla \widetilde{h}_{t-d}(\widehat{\mathbf{\Phi}}(t-d)) \right], \quad d \leq t < T. \quad \text{(S.10)}$$

For a communication graph, let $\mathcal{N}_\kappa(i)$ be the sources within κ directed hops of receiver i. Define

$$\mathcal{F}_{d,\kappa} = \left\{ \widehat{\mathbf{\Phi}} \in \mathcal{F}_d : [\widehat{\Phi}^a_{\ell,k}]_{i,j} = 0 \text{ for } j \notin \mathcal{N}_\kappa(i) \right\}.$$

The localized loss $\widetilde{h}^\kappa_s$ replaces $\widehat{\mathbf{\Phi}}^{\mathrm{off}}$ in both terms of (S.6) by $\widehat{\mathbf{\Phi}}^{\mathrm{off},\kappa} \in \mathcal{F}_{d,\kappa}$. Its receiver-$i$ residual and gradient are

$$r^{a,i}_\ell(\widehat{\mathbf{\Phi}}, \widehat{\mathbf{w}}) := \sum_{k \in \mathcal{I}_\ell} \sum_{j \in \mathcal{N}_\kappa(i)} \left([\widehat{\Phi}^a_{\ell,k}]_{i,j} [\widehat{\mathbf{w}}]^j_k - [\widehat{\Phi}^{a,\mathrm{off},\kappa}_{\ell,k}]_{i,j} [\mathbf{w}]^j_k \right), \quad \text{(S.11)}$$

$$\left[\nabla_{\widehat{\Phi}^a_{\ell,k}} \widetilde{h}^{i,\kappa}_s \right]_{i,j} = r^{a,i}_\ell(\widehat{\mathbf{\Phi}}, \widehat{\mathbf{w}}) ([\widehat{\mathbf{w}}]^j_k)^\top + \lambda [\widehat{\Phi}^a_{\ell,k} - \widehat{\Phi}^{a,\mathrm{off},\kappa}_{\ell,k}]_{i,j}, \quad \text{(S.12)}$$

for $\ell \leq s$, $k \in \mathcal{I}_\ell$, and $j \in \mathcal{N}_\kappa(i)$. Here $\widetilde{h}^{i,\kappa}_s$ collects the state and input rows of receiver i from $\widetilde{h}^\kappa_s$. The localized hindsight response is

$$\widehat{\mathbf{\Phi}}^{\diamond,\kappa} := \arg\min_{\widehat{\mathbf{\Phi}} \in \mathcal{F}_{d,\kappa}} \widetilde{h}^\kappa_T(\widehat{\mathbf{\Phi}}). \quad \text{(S.13)}$$

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Let δ_{ji} be the directed hop distance from j to i and set

$$\delta_i = \max_{j\in\mathcal{N}_\kappa(i)} \delta_{ji}, \qquad D_i = d + \delta_i.$$

Write $[\widehat{\mathbf{\Phi}}]_i$ for the state and input response rows of receiver i, and $\mathcal{F}^i_{d,\kappa}$ for their support constraint. With a local step size $\eta_i > 0$, initialize $[\widehat{\mathbf{\Phi}}]_i(t) = [\widehat{\mathbf{\Phi}}^{\mathrm{off},\kappa}]_i$ for $0 \le t \le D_i$ and use

$$[\widehat{\mathbf{\Phi}}]_i(t+1) = \Pi_{\mathcal{F}^i_{d,\kappa}}\Big[[\widehat{\mathbf{\Phi}}]_i(t) - \eta_i \nabla \widetilde{h}^{i,\kappa}_{t-D_i}([\widehat{\mathbf{\Phi}}]_i(t-D_i))\Big],$$

Appendix A
Proof of Lemma 2

Proof. As the first step, we introduce the following cost difference lemma

Lemma S.1 (Cost difference lemma)**.** *Given the stabilizable system A, B, Q, R and control policy π, the difference between the offline optimal trajectory $(x_0^\star, u_0^\star, x_1^\star, \dots)$ and the one using $\boldsymbol{\pi}$ satisfies*

$$\mathtt{Reg}_T(\boldsymbol{\pi}) = \sum_{t=0}^{T-1} \left\| V\left(u_t^\pi - u_t^\star\right) + B^\top P A \left(x_{t+1}^\pi - x_{t+1}^\star\right)\right\|^2_{V^{-1}},$$

where $V \coloneqq R + B^\top P B$; $\|\cdot\|^2_{V^{-1}} \coloneqq (\cdot)^\top V^{-1}(\cdot)$ denotes the generalized ℓ_2-norm as a quadratic form and P is the solution of DARE.

We omit the proof for Lemma S.1, since it can be done by simple algebraic manipulation from Lemma 13 in [1]. Given the quadratic cost (S.2), we have:

$$\begin{aligned}
&\mathtt{Reg}_T(\mathbf{\Phi}; \widehat{\mathbf{w}}) = J_T(\boldsymbol{\pi}; \widehat{\mathbf{w}}_{0:T-1}) - J_T^\star \\
&= \sum_{\ell=0}^{T-1} \left\| V\left(u_\ell(\mathbf{\Phi}; \widehat{\mathbf{w}}) - u_\ell^\star\right) + B^\top P A \left(x_\ell(\mathbf{\Phi}; \widehat{\mathbf{w}}) - x_\ell^\star\right)\right\|^2_{V^{-1}} \\
&\le \sum_{\ell=0}^{T-1} 2\left\| V\left(u_\ell(\mathbf{\Phi}; \widehat{\mathbf{w}}) - u_\ell^\star\right)\right\|^2_{V^{-1}} + 2\left\| B^\top P A \left(x_\ell(\mathbf{\Phi}; \widehat{\mathbf{w}}) - x_\ell^\star\right)\right\|^2_{V^{-1}}
\end{aligned}$$

Given CLMs, the formulation can be upper bounded by

$$\begin{aligned}
&2\|V^{-1}\|_2^2 \sum_{\ell=0}^{T-1} \left\| \sum_{k=\underline{\ell}}^{\overline{\ell}} \left(\Phi^u_{\ell,k} w_{k-1} + \widehat{\Phi}^u_{\ell,k} \widehat{w}_{k-1}\right) \right. \\
&\qquad \left. - \sum_{k=0}^{T} \left(\Phi^{u\mathrm{off}}_{\ell,k} w_{k-1} + \widehat{\Phi}^{u\mathrm{off}}_{\ell,k} w_{k-1}\right)\right\|^2 \\
&\quad + 2\|H\|_2^2 \sum_{\ell=0}^{T-1} \left\| \sum_{k=\underline{\ell}}^{\overline{\ell}} \left(\Phi^x_{\ell,k} w_{k-1} + \widehat{\Phi}^x_{\ell,k} \widehat{w}_{k-1}\right) \right. \\
&\qquad \left. - \sum_{k=0}^{T} \left(\Phi^{x\mathrm{off}}_{\ell,k} w_{k-1} + \widehat{\Phi}^{x\mathrm{off}}_{\ell,k} w_{k-1}\right)\right\|^2
\end{aligned}$$

Separate causal and non-causal CLMs to obtain the following:

$$\begin{aligned}
&4\omega(\|H\|_2^2 \left\|\mathbf{\Phi}^x - \mathbf{\Phi}^{x\mathrm{off}}\right\|_{\mathcal{F}} + \|V^{-1}\|_2^2 \left\|\mathbf{\Phi}^u - \mathbf{\Phi}^{u\mathrm{off}}\right\|_{\mathcal{F}}) \\
&\quad + 4\left(\|H\|_2^2 \sum_{\ell=0}^{T-1} \left\| \sum_{k=\underline{\ell}}^{\overline{\ell}} \left(\widehat{\Phi}^x_{\ell,k} \widehat{w}_{k-1} - \widehat{\Phi}^{x\mathrm{off}}_{\ell,k} w_{k-1}\right)\right\|^2 \right. \\
&\quad \left. + \|V^{-1}\|_2^2 \sum_{\ell=0}^{T-1} \left\| \sum_{k=\underline{\ell}}^{\overline{\ell}} \left(\widehat{\Phi}^u_{\ell,k} \widehat{w}_{k-1} - \widehat{\Phi}^{u\mathrm{off}}_{\ell,k} w_{k-1}\right)\right\|^2 \right) \\
&\quad \underbrace{+ 4\omega \max(\|H\|_2^2, \|V^{-1}\|_2^2) \sum_{a\in\{x,u\}} \sum_{\ell=0}^{T-1} \sum_{|k-\ell|\ge d} \left\|\Phi^{a\mathrm{off}}_{\ell,k} + \widehat{\Phi}^{a\mathrm{off}}_{\ell,k}\right\|^2_{\mathcal{F}}}_{\text{(a)}}
\end{aligned}$$

where $H \coloneqq B^\top P A V^{-1} A^\top P B$ and both norms of V^{-1} and H can be upper bounded by the stabilizable system. (a) is an unavoidable extra item, which can be further upper bounded by:

$$\begin{aligned}
\text{(a)} &\le 8\phi\omega \sum_{\ell=0}^{T-1} \sum_{k=d}^{\infty} \varrho^k \cdot \max\left(\|H\|_2^2, \|V^{-1}\|_2^2\right) \\
&\le \frac{8\phi\omega}{1-\varrho} \cdot \max\left(\|H\|_2^2, \|V^{-1}\|_2^2\right) \cdot T\varrho^d = O(T\varrho^d)
\end{aligned}$$

here the first inequality employs the following lemma:

Lemma S.2 (Decaying property, Theorem 5.1 in [1])**.** *Denote by $(\Phi^{x\mathrm{off}}_{\ell,k}, \Phi^{u\mathrm{off}}_{\ell,k}, \widehat{\Phi}^{x\mathrm{off}}_{\ell,k}, \widehat{\Phi}^{u\mathrm{off}}_{\ell,k})$ at time indeces $\ell, k \in [T]$ an optimal solution in Lemma 1. Under the model assumption, there exist constants $\phi > 0$ and $\rho \in (0,1)$ dependent on the system matrices (A, B, Q, R) in* (S.1) *such that* $\max(\|\Phi^{x\mathrm{off}}_{t,k}\|^2, \|\widehat{\Phi}^{x\mathrm{off}}_{t,k}\|^2, \|\Phi^{u\mathrm{off}}_{t,k}\|^2, \|\widehat{\Phi}^{u\mathrm{off}}_{t,k}\|^2) \le \phi\varrho^{|t-k|}$.

this shows the truncation error incurred by restricting (S.4) to the space of FIR CLMs decays exponentially with d. □

Appendix B
Proof of Theorem 1

Proof. Fix $\ell \in [T]$ and let $\Delta_\ell \mathbf{\Phi}(t) \coloneqq \mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}^\diamond_{\ell,:}, t_\ell \coloneqq \max\{2d, d+\ell\}$.

Step 1: Establish the one-step recursive relation:

Since $\widetilde{h}_s$ is separable with respect to its block rows, for every t, let $\left[\widetilde{\mathbf{\Psi}}(\mathbf{\Phi}(t))\right]_{\ell,:} = \nabla_{\mathbf{\Phi}_{\ell,:}} \widetilde{h}_\ell(\mathbf{\Phi}(q))$. Moreover, the optimality of $\mathbf{\Phi}^\diamond$ gives

$$\left[\widetilde{\mathbf{\Psi}}(\mathbf{\Phi}^\diamond)\right]_{\ell,:} = \nabla_{\mathbf{\Phi}_{\ell,:}} \widetilde{h}_\ell(\mathbf{\Phi}^\diamond) = 0.$$

Therefore, Algorithm 1 gives

$$\Delta\mathbf{\Phi}_\ell(t+1) = \Delta\mathbf{\Phi}_\ell(t) - \eta\left[\widetilde{\mathbf{\Psi}}(\mathbf{\Phi}(\underline{q})) - \widetilde{\mathbf{\Psi}}(\mathbf{\Phi}^\diamond)\right]_{\ell,:}. \quad \text{(S.14)}$$

For clarity, define the row-gradient differences

$$\mathbf{g}_\ell(t) \coloneqq \left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(t)\big) - \widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}^\diamond\big)\right]_{\ell,:},$$

Expanding the squared Frobenius norm in (S.14) yields

$$\begin{aligned}
\|\Delta\mathbf{\Phi}_\ell(t+1)\|^2_{\mathcal{F}} \le \|\Delta\mathbf{\Phi}_\ell(t)\|^2_{\mathcal{F}} &- 2\eta\left\langle \mathbf{g}_\ell(\underline{t}), \Delta\mathbf{\Phi}_\ell(t)\right\rangle \\
&+ \eta^2 \|\mathbf{g}_\ell(\underline{t})\|^2_{\mathcal{F}}. \quad \text{(S.15)}
\end{aligned}$$

We now add and subtract the current row gradient $\mathbf{g}_\ell(t)$ in the inner-product term:

$$-2\eta\,\langle \mathbf{g}_\ell(\underline{t}), \Delta\mathbf{\Phi}_\ell(t)\rangle = -2\eta\,\langle \mathbf{g}_\ell(t), \Delta\mathbf{\Phi}_\ell(t)\rangle + 2\eta\,\langle \mathbf{g}_\ell(t) - \mathbf{g}_\ell(\underline{t}), \Delta\mathbf{\Phi}_\ell(t)\rangle\,. \tag{S.16}$$

Because the ℓ-th row term of the surrogate is λ-strongly convex, its gradient is λ-strongly monotone. Therefore,

$$\langle \mathbf{g}_\ell(t), \Delta\mathbf{\Phi}_\ell(t)\rangle = \left\langle \left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(t)\big) - \widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}^\diamond\big)\right]_{\ell,:}, \mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}^\diamond_{\ell,:}\right\rangle \geq \lambda\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2\,. \tag{S.17}$$

Substituting (S.16) and (S.17) into (S.15), we obtain

$$\|\Delta\mathbf{\Phi}_\ell(t+1)\|_{\mathcal{F}}^2 \leq (1-2\lambda\eta)\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2 + \eta^2\,\|\mathbf{g}_\ell(\underline{t})\|_{\mathcal{F}}^2 + 2\eta\,\langle \mathbf{g}_\ell(t) - \mathbf{g}_\ell(\underline{t}), \Delta\mathbf{\Phi}_\ell(t)\rangle\,. \tag{S.18}$$

Finally, since $\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}^\diamond\big)\right]_{\ell,:} = 0$, we have $\mathbf{g}_\ell(\underline{t}) = \left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}$, and $\mathbf{g}_\ell(t) - \mathbf{g}_\ell(\underline{t}) = \left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(t)\big) - \widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}$. Hence (S.18) becomes

$$\|\Delta\mathbf{\Phi}_\ell(t+1)\|_{\mathcal{F}}^2 \leq (1-2\lambda\eta)\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2 + \eta^2\left\|\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}\right\|_{\mathcal{F}}^2 + 2\eta\left\langle \left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(t)\big) - \widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}, \Delta\mathbf{\Phi}_\ell(t)\right\rangle.$$

By the L-smoothness of $\widetilde{h}_\ell$,

$$\left\|\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(t)\big) - \widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}\right\|_{\mathcal{F}} \leq L\,\|\mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}_{\ell,:}(\underline{t})\|_{\mathcal{F}}\,. \tag{S.19}$$

Applying Cauchy–Schwarz and Young's inequality gives

$$\|\Delta\mathbf{\Phi}_\ell(t+1)\|_{\mathcal{F}}^2 \leq (1-\lambda\eta)\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2 + \eta^2\left\|\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}\right\|_{\mathcal{F}}^2 + \frac{\eta L^2}{\lambda}\,\|\mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}_{\ell,:}(\underline{t})\|_{\mathcal{F}}^2\,. \tag{S.20}$$

Since $\underline{t} = t-d$, telescoping the last d updates gives

$$\|\mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}_{\ell,:}(\underline{t})\|_{\mathcal{F}} \leq \sum_{k=1}^{d}\|\mathbf{\Phi}_{\ell,:}(t-k+1) - \mathbf{\Phi}_{\ell,:}(t-k)\|_{\mathcal{F}} \leq \eta\sum_{k=1}^{d}\left\|\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(t-k-d)\big)\right]_{\ell,:}\right\|_{\mathcal{F}} \leq \eta L\sum_{k=1}^{d}\|\Delta\mathbf{\Phi}_\ell(t-k-d)\|_{\mathcal{F}}\,. \tag{S.21}$$

Consequently,

$$\|\mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}_{\ell,:}(\underline{t})\|_{\mathcal{F}}^2 \leq d\eta^2L^2\sum_{k=1}^{d}\|\Delta\mathbf{\Phi}_\ell(t-k-d)\|_{\mathcal{F}}^2\,. \tag{S.22}$$

Likewise, using $\widetilde{\mathbf{\Psi}}(\mathbf{\Phi}^\diamond) = 0$, (S.19), and (S.21), we obtain

$$\left\|\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}\right\|_{\mathcal{F}} \leq L\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}} + L\,\|\mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}_{\ell,:}(\underline{t})\|_{\mathcal{F}} \leq L\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}} + \eta L^2\sum_{k=1}^{d}\|\Delta\mathbf{\Phi}_\ell(t-k-d)\|_{\mathcal{F}}\,. \tag{S.23}$$

Squaring (S.23) yields

$$\left\|\left[\widetilde{\mathbf{\Psi}}\big(\mathbf{\Phi}(\underline{t})\big)\right]_{\ell,:}\right\|_{\mathcal{F}}^2 \leq 2L^2\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2 + 2d\eta^2L^4\sum_{k=1}^{d}\|\Delta\mathbf{\Phi}_\ell(t-k-d)\|_{\mathcal{F}}^2\,. \tag{S.24}$$

Substituting (S.22) and (S.24) into (S.20), we have

$$\|\Delta\mathbf{\Phi}_\ell(t+1)\|_{\mathcal{F}}^2 \leq c_0(\eta)\,\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2 + c_1(\eta)\sum_{k=1}^{d}\|\Delta\mathbf{\Phi}_\ell(t-k-d)\|_{\mathcal{F}}^2\,, \tag{S.25}$$

where $c_0(\eta) := 1-\lambda\eta+2\eta^2L^2$, and $c_1(\eta) := 2d\eta^4L^4 + \frac{d\eta^3L^4}{\lambda}$.

Step 2: Establish contraction of the delayed error window: By definition,

$$c_0(\eta)+dc_1(\eta) = 1-\lambda\eta+2\eta^2L^2+\frac{d^2\eta^3L^4}{\lambda}+2d^2\eta^4L^4 = \rho(\lambda).$$

The stepsize conditions imply $0 < \rho(\lambda) \leq 1 - \frac{\lambda\eta}{2} < 1$. For $t \geq t_\ell$, define

$$\widetilde{\delta}_\ell(t) := \max_{0\leq k\leq 2d}\|\Delta\mathbf{\Phi}_\ell(t-k)\|_{\mathcal{F}}^2\,.$$

Using (S.25), we obtain

$$\|\Delta\mathbf{\Phi}_\ell(t+1)\|_{\mathcal{F}}^2 \leq \big(c_0(\eta)+dc_1(\eta)\big)\widetilde{\delta}_\ell(t) = \rho(\lambda)\widetilde{\delta}_\ell(t). \tag{S.26}$$

Since $\rho(\lambda) < 1$, we have

$$\widetilde{\delta}_\ell(t+1) \leq \widetilde{\delta}_\ell(t).$$

Therefore, for every $j = 1, \ldots, 2d+1$,

$$\|\Delta\mathbf{\Phi}_\ell(t+j)\|_{\mathcal{F}}^2 \leq \rho(\lambda)\widetilde{\delta}_\ell(t),$$

and consequently

$$\widetilde{\delta}_\ell(t+2d+1) \leq \rho(\lambda)\widetilde{\delta}_\ell(t).$$

By induction,

$$\widetilde{\delta}_\ell(t) \leq \rho(\lambda)^{\left\lfloor\frac{t-t_\ell}{2d+1}\right\rfloor}\widetilde{\delta}_\ell(t_\ell), \qquad t \geq t_\ell. \tag{S.27}$$

It remains to bound the initial window. If $\ell \geq d$, then $t_\ell = \ell + d$, and row ℓ has not been updated before t_ℓ. Hence

$$\widetilde{\delta}_\ell(t_\ell) = \left\|\mathbf{\Phi}^{\text{off}}_{\ell,:} - \mathbf{\Phi}^\diamond_{\ell,:}\right\|_{\mathcal{F}}^2.$$

If $\ell < d$, then $t_\ell = 2d$. Before time $2d$, every delayed argument used to update row ℓ is still equal to its offline initialization. Since the row objective is quadratic, for $0 \leq j \leq d-\ell$,

$$\Delta\mathbf{\Phi}_\ell(\ell+d+j) = (I - j\eta H_\ell)\Delta\mathbf{\Phi}^{\text{off}}_\ell,$$

where $\lambda I \preceq H_\ell \preceq LI$. The stepsize condition gives $j\eta L \le d\eta L \le \frac{\lambda}{\sqrt{6}L} < 1$, and thus

$$\|\Delta\mathbf{\Phi}_\ell(\ell+d+j)\|_{\mathcal{F}} \le \left\|\Delta\mathbf{\Phi}_\ell^{\mathrm{off}}\right\|_{\mathcal{F}}.$$

Therefore, in both cases,

$$\widetilde{\delta}_\ell(t_\ell) \le \left\|\mathbf{\Phi}_{\ell,:}^{\mathrm{off}} - \mathbf{\Phi}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2.$$

Finally, since $\|\Delta\mathbf{\Phi}_\ell(t)\|_{\mathcal{F}}^2 \le \widetilde{\delta}_\ell(t)$, (S.27) gives

$$\left\|\mathbf{\Phi}_{\ell,:}(t) - \mathbf{\Phi}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2 \le \rho(\lambda)^{\left\lfloor \frac{t-\max\{2d,d+\ell\}}{2d+1}\right\rfloor} \left\|\mathbf{\Phi}_{\ell,:}^{\mathrm{off}} - \mathbf{\Phi}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2.$$

This proves the convergence result. □

Appendix C
Proof of Proposition 2

To prove the dynamic regret result, we first show a result about the gap between $J(\mathbf{\Phi}(T);\widehat{\mathbf{w}})$ and hindsight optimal performance $J(\mathbf{\Phi}^{\diamond};\widehat{\mathbf{w}})$.

Lemma S.3 (Hindsight Regret)**.** *Under the conditions of Theorem 1, for every $\upsilon > 0$ and $T \ge 2d$,*

$$\begin{aligned} &J\big(\widehat{\mathbf{\Phi}}(T);\widehat{\mathbf{w}}\big) - (1+\upsilon)J\big(\widehat{\mathbf{\Phi}}^{\diamond};\widehat{\mathbf{w}}\big) \\ &\le \left(1+\frac{1}{\upsilon}\right) O\left(\|\widehat{\mathbf{w}}\|_2^2\left[4d + \frac{(2d+1)\rho(\lambda)}{1-\rho(\lambda)}\right]\right), \end{aligned}$$

where $\rho(\lambda) \in (0,1)$ is given in Theorem 1.

Proof of Lemma S.3. Since $\widehat{\mathbf{\Phi}}(T)$ and $\widehat{\mathbf{\Phi}}^{\diamond}$ are fixed CLMs. Therefore,

$$\begin{aligned} &\sum_{t=0}^{T-1}\left(\|x_t - x_t^{\diamond}\|_Q^2 + \|u_t - u_t^{\diamond}\|_R^2\right) + \|x_T - x_T^{\diamond}\|_P^2 \\ &\quad\le \|\widehat{\mathbf{w}}\|_2^2 \max\{\|Q\|,\|R\|,\|P\|\}\left\|\widehat{\mathbf{\Phi}}(T) - \widehat{\mathbf{\Phi}}^{\diamond}\right\|_{\mathcal{F}}^2. \quad \text{(S.28)} \end{aligned}$$

Decomposing the Frobenius norm across the block rows gives

$$\begin{aligned} \left\|\widehat{\mathbf{\Phi}}(T) - \widehat{\mathbf{\Phi}}^{\diamond}\right\|_{\mathcal{F}}^2 &= \sum_{\ell=0}^{T-d-1}\left\|\widehat{\mathbf{\Phi}}_{\ell,:}(T) - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2 \\ &\quad + \sum_{\ell=T-d}^{T-1}\left\|\widehat{\mathbf{\Phi}}_{\ell,:}(T) - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2. \quad \text{(S.29)} \end{aligned}$$

By the assumed uniform boundedness of the rowwise terminal perturbation, the second sum contains at most d uniformly bounded terms and therefore satisfies

$$\sum_{\ell=T-d}^{T-1}\left\|\widehat{\mathbf{\Phi}}_{\ell,:}(T) - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2 = O(d). \quad \text{(S.30)}$$

Therefore, applying Theorem 1 at $t = T$ gives

$$\left\|\widehat{\mathbf{\Phi}}_{\ell,:}(T) - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2 \le \rho(\lambda)^{\left\lfloor \frac{T-\max\{2d,d+\ell\}}{2d+1}\right\rfloor}\left\|\widehat{\mathbf{\Phi}}_{\ell,:}^{\mathrm{off}} - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2.$$

Summing the above bound over $\ell = 0,\dots,T-d-1$ and using (S.29)–(S.30) yield

$$\begin{aligned} &\left\|\widehat{\mathbf{\Phi}}(T) - \widehat{\mathbf{\Phi}}^{\diamond}\right\|_{\mathcal{F}}^2 \le \\ &\sum_{\ell=0}^{T-d-1}\rho(\lambda)^{\left\lfloor \frac{T-\max\{2d,d+\ell\}}{2d+1}\right\rfloor}\left\|\widehat{\mathbf{\Phi}}_{\ell,:}^{\mathrm{off}} - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2 + O(d). \quad \text{(S.31)} \end{aligned}$$

Finally, for every $\upsilon > 0$, Young's inequality gives

$$\begin{aligned} &J\big(\widehat{\mathbf{\Phi}}(T);\widehat{\mathbf{w}}\big) - (1+\upsilon)J\big(\widehat{\mathbf{\Phi}}^{\diamond};\widehat{\mathbf{w}}\big) \le \left(1+\frac{1}{\upsilon}\right) \quad \text{(S.32)} \\ &\times\left[\sum_{t=0}^{T-1}\left(\|x_t - x_t^{\diamond}\|_Q^2 + \|u_t - u_t^{\diamond}\|_R^2\right) + \|x_T - x_T^{\diamond}\|_P^2\right]. \end{aligned}$$

Combining (S.32), (S.28), and (S.31) proves

$$\begin{aligned} &J\big(\widehat{\mathbf{\Phi}}(T);\widehat{\mathbf{w}}\big) - (1+\upsilon)J\big(\widehat{\mathbf{\Phi}}^{\diamond};\widehat{\mathbf{w}}\big) \\ &\le \left(1+\frac{1}{\upsilon}\right)\|\widehat{\mathbf{w}}\|_2^2 \max\{\|Q\|,\|R\|,\|P\|\}\Bigg[\\ &\quad\sum_{\ell=0}^{T-d-1}\rho(\lambda)^{\left\lfloor \frac{T-\max\{2d,d+\ell\}}{2d+1}\right\rfloor}\left\|\widehat{\mathbf{\Phi}}_{\ell,:}^{\mathrm{off}} - \widehat{\mathbf{\Phi}}_{\ell,:}^{\diamond}\right\|_{\mathcal{F}}^2 + O(d)\Bigg], \end{aligned}$$

For $\ell = 0,\dots,d-1$, $\max\{2d, d+\ell\} = 2d$, whereas, for $\ell = d,\dots,T-d-1$, $\max\{2d,d+\ell\} = d+\ell$. Therefore,

$$\begin{aligned} &\sum_{\ell=0}^{T-d-1}\rho(\lambda)^{\left\lfloor \frac{T-\max\{2d,d+\ell\}}{2d+1}\right\rfloor} \\ &= \sum_{\ell=0}^{d-1}\rho(\lambda)^{\left\lfloor \frac{T-2d}{2d+1}\right\rfloor} + \sum_{\ell=d}^{T-d-1}\rho(\lambda)^{\left\lfloor \frac{T-d-\ell}{2d+1}\right\rfloor} \\ &= d\,\rho(\lambda)^{\left\lfloor \frac{T-2d}{2d+1}\right\rfloor} + \sum_{j=1}^{T-2d}\rho(\lambda)^{\left\lfloor \frac{j}{2d+1}\right\rfloor}, \quad \text{(S.33)} \end{aligned}$$

where the last equality follows from the change of index $j = T-d-\ell$. Let $m = \left\lfloor \frac{T-2d}{2d+1}\right\rfloor$. For $j = 1,\dots,T-2d$, the exponent zero occurs for at most $2d$ indices. For each $k \in \{1,\dots,m\}$, the set $\left\{j : \left\lfloor \frac{j}{2d+1}\right\rfloor = k\right\}$ contains at most $2d+1$ indices. Consequently,

$$\begin{aligned} \sum_{j=1}^{T-2d}\rho(\lambda)^{\left\lfloor \frac{j}{2d+1}\right\rfloor} &\le 2d + (2d+1)\sum_{k=1}^{m}\rho(\lambda)^k \\ &= 2d + (2d+1)\frac{\rho(\lambda)(1-\rho(\lambda)^m)}{1-\rho(\lambda)}. \quad \text{(S.34)} \end{aligned}$$

Because $\rho(\lambda)^m \le 1$, combining (S.33) and (S.34) yields

$$\sum_{\ell=0}^{T-d-1}\rho(\lambda)^{\left\lfloor \frac{T-\max\{2d,d+\ell\}}{2d+1}\right\rfloor} \le 3d + \frac{(2d+1)\rho(\lambda)}{1-\rho(\lambda)}, \quad \text{(S.35)}$$

which completes the proof. □

Proof of Theorem 2. Recall that the definition of the hindsight optimal policy is as

$$\mathbf{\Phi}^{\diamond} = \operatorname*{arg\,min}_{\mathbf{\Phi}\in\mathcal{F}_T}\widetilde{h}_T(\mathbf{\Phi};\widehat{\mathbf{w}},\mathbf{w}), \quad \text{(S.36)}$$

This minimizer is well-defined since the strong convexity of $\widetilde{h}_T(\mathbf{\Phi};\widehat{\mathbf{w}},\mathbf{w})$ in $\mathbf{\Phi}$ guarantees that $\mathbf{\Phi}^{\diamond}$ is unique and can

be explicitly written out by a system-level gradient:

$$\begin{aligned}
&\nabla_{\mathbf{\Phi}} \widetilde{h}_T(\mathbf{\Phi}; \widehat{\mathbf{w}}) = 0 \\
\Rightarrow &\nabla_{\mathbf{\Phi}} \left(\frac{1}{2} \|\widehat{\mathbf{\Phi}}^x \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{\mathrm{xoff}} \mathbf{w}\|^2 + \frac{1}{2} \|\widehat{\mathbf{\Phi}}^{\mathrm{u}} \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{\mathrm{uoff}} \mathbf{w}\|^2 \right. \\
&\left. + \frac{\lambda}{2} \|\widehat{\mathbf{\Phi}}^x - \widehat{\mathbf{\Phi}}^{\mathrm{xoff}}\|^2 + \frac{\lambda}{2} \|\widehat{\mathbf{\Phi}}^{\mathrm{u}} - \widehat{\mathbf{\Phi}}^{\mathrm{uoff}}\|^2 \right) = 0 \\
\Rightarrow &\begin{bmatrix} (\widehat{\mathbf{\Phi}}^x \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{\mathrm{xoff}} \mathbf{w}) \widehat{\mathbf{w}}^\top + \lambda (\widehat{\mathbf{\Phi}}^x - \widehat{\mathbf{\Phi}}^{\mathrm{xoff}}) \\ (\widehat{\mathbf{\Phi}}^{\mathrm{u}} \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{\mathrm{uoff}} \mathbf{w}) \widehat{\mathbf{w}}^\top + \lambda (\widehat{\mathbf{\Phi}}^{\mathrm{u}} - \widehat{\mathbf{\Phi}}^{\mathrm{uoff}}) \end{bmatrix} = 0 \\
\Rightarrow &\begin{bmatrix} \widehat{\mathbf{\Phi}}^x \\ \widehat{\mathbf{\Phi}}^{\mathrm{u}} \end{bmatrix} = \begin{bmatrix} \widehat{\mathbf{\Phi}}^{\mathrm{xoff}} (\mathbf{w} \widehat{\mathbf{w}}^\top + \lambda I)(\widehat{\mathbf{w}} \widehat{\mathbf{w}}^\top + \lambda I)^{-1} \\ \widehat{\mathbf{\Phi}}^{\mathrm{uoff}} (\mathbf{w} \widehat{\mathbf{w}}^\top + \lambda I)(\widehat{\mathbf{w}} \widehat{\mathbf{w}}^\top + \lambda I)^{-1} \end{bmatrix}
\end{aligned}$$

Using this equality, we obtain

$$\begin{aligned}
&\sum_{t=0}^{T-1} \left\| x_{t+1}^{\diamond} - x_{t+1}^{\star(d)} \right\|^2 = \left\| \mathbf{x}^{\diamond} - \mathbf{x}^{\star(d)} \right\|^2 = \left\| \widehat{\mathbf{\Phi}}^x \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{x\mathrm{off}} \mathbf{w} \right\|^2 \\
&= \left\| \widehat{\mathbf{\Phi}}^{x\mathrm{off}} (\mathbf{w} \widehat{\mathbf{w}}^\top + \lambda I)(\widehat{\mathbf{w}} \widehat{\mathbf{w}}^\top + \lambda I)^{-1} \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{x\mathrm{off}} \mathbf{w} \right\|^2 \\
&= \left\| \widehat{\mathbf{\Phi}}^{x\mathrm{off}} (\mathbf{w} \widehat{\mathbf{w}}^\top + \lambda I) \left(\frac{1}{\lambda} I - \frac{\widehat{\mathbf{w}} \widehat{\mathbf{w}}^\top}{\lambda (\lambda + \|\widehat{\mathbf{w}}\|^2)} \right) \widehat{\mathbf{w}} - \widehat{\mathbf{\Phi}}^{x\mathrm{off}} \mathbf{w} \right\|^2 \\
&= \left\| \widehat{\mathbf{\Phi}}^{x\mathrm{off}} \left(\frac{\|\widehat{\mathbf{w}}\|^2 \mathbf{w} + \lambda \widehat{\mathbf{w}}}{\lambda + \|\widehat{\mathbf{w}}\|^2} - \mathbf{w} \right) \right\|^2 \\
&= \left(\frac{\lambda}{\lambda + \|\widehat{\mathbf{w}}\|^2} \right)^2 \left\| \widehat{\mathbf{\Phi}}^{x\mathrm{off}} (\widehat{\mathbf{w}} - \mathbf{w}) \right\|^2
\end{aligned}$$

where $x_t^{\star(d)} := \sum_{k=\underline{t}}^{t} \Phi_{\ell,k}^{x\mathrm{off}} \mathbf{w}_{k-1} + \sum_{k=\underline{t}}^{\bar{t}} \widehat{\Phi}_{\ell,k}^{x\mathrm{off}} \mathbf{w}_{k-1}$ denoting the FIR truncated version of offline optimal state at time step t; the forth equality utilizes the Sherman-Morrison formula, and we can obtain $\sum_{t=0}^{T-1} \left\| u_t^{\diamond} - u_t^{\star(d)} \right\|^2 = \left(\frac{\lambda}{\lambda + \|\widehat{\mathbf{w}}\|^2} \right)^2 \left\| \widehat{\mathbf{\Phi}}^{x\mathrm{off}} (\widehat{\mathbf{w}} - \mathbf{w}) \right\|^2$ using the identical method.

Now it suffices to obtain the bound of cost function differences. Recall the inequality presented in Lemma S.1, we have

$$\begin{aligned}
&J\big(\mathbf{\Phi}^{\diamond}; \widehat{\mathbf{w}}\big) - J^{\star} = \sum_{t=0}^{T-1} \left\| V \left(u_t^{\diamond} - u_t^{\star} \right) + B^\top P A \left(x_{t+1}^{\diamond} - x_{t+1}^{\star} \right) \right\|_{V^{-1}}^2 \\
&\leq \sum_{t=0}^{T-1} 2 \left\| V \left(u_t^{\diamond} - u_t^{\star} \right) \right\|_{V^{-1}}^2 + 2 \left\| B^\top P A \left(x_{t+1}^{\diamond} - x_{t+1}^{\star} \right) \right\|_{V^{-1}}^2 \\
&\leq 2 \|V^{-1}\|_2^2 \sum_{t=0}^{T-1} \|u_t^{\diamond} - u_t^{\star}\|^2 + 2 \|H\|_2^2 \sum_{t=0}^{T-1} \|x_{t+1}^{\diamond} - x_{t+1}^{\star}\|^2 \\
&\leq 4 \|V^{-1}\|_2^2 \sum_{t=0}^{T-1} \left\| u_t^{\diamond} - u_t^{\star(d)} \right\|^2 + 4 \|H\|_2^2 \sum_{t=0}^{T-1} \left\| x_{t+1}^{\diamond} - x_{t+1}^{\star(d)} \right\|^2 \\
&\quad + \frac{4 \phi \omega \max\left(\|H\|_2^2, \|V^{-1}\|_2^2 \right)}{1 - \varrho} \cdot T \varrho^d \\
&\leq 4 \max\left(\|H\|_2^2, \|V^{-1}\|_2^2 \right) \cdot \left(\frac{\lambda}{\lambda + \|\widehat{\mathbf{w}}\|^2} \right)^2 \varepsilon \\
&\quad + \frac{4 \phi \omega \max\left(\|H\|_2^2, \|V^{-1}\|_2^2 \right)}{1 - \varrho} \cdot T \varrho^d
\end{aligned}$$

where $H := B^\top P A V^{-1} A^\top P B$; the third inequality employs the temporally decaying property of offline PredSLS CLMs, and the forth inequality employs the above equation.

Applying Lemma S.3, it now suffices to prove the regret in Proposition 2. Consider the difference of the costs which can be decomposed as:

$$\begin{aligned}
&J\big(\mathbf{\Phi}(T); \widehat{\mathbf{w}}\big) - (1 + \upsilon) J^{\star} \\
&= J\big(\mathbf{\Phi}(T); \widehat{\mathbf{w}}\big) - (1 + \upsilon) J\big(\mathbf{\Phi}^{\diamond}; \widehat{\mathbf{w}}\big) + (1 + \upsilon) \left(J\big(\mathbf{\Phi}^{\diamond}; \widehat{\mathbf{w}}\big) - J^{\star} \right) \\
&\leq \left(1 + \frac{1}{\upsilon} \right) O\left(\|\widehat{\mathbf{w}}\|_2^2 \left[4d + \frac{(2d+1)\rho(\lambda)}{1 - \rho(\lambda)} \right] \right) \\
&\quad + (1 + \upsilon) \left(\max\left(\|H\|_2^2, \|V^{-1}\|_2^2 \right) \cdot \left(\frac{\lambda}{\lambda + \|\widehat{\mathbf{w}}\|^2} \right)^2 \varepsilon \right. \\
&\quad \left. + \frac{4 \phi \omega \max\left(\|H\|_2^2, \|V^{-1}\|_2^2 \right)}{1 - \varrho} \cdot T \varrho^d \right) \\
&= \left(1 + \frac{1}{\upsilon} \right) \widetilde{O}\left(T - (1 - e^{\frac{\max\{\tau, d\}}{6}} \rho(\lambda)^{\frac{T - \max\{\tau, d\}}{d+1}}) \tau \right) \\
&\quad + (1 + \upsilon) O\left(\left(\frac{\lambda}{\lambda + \|\widehat{\mathbf{w}}\|^2} \right)^2 \varepsilon \right) + (1 + \upsilon) O(T \varrho^d)
\end{aligned}$$

To bound the optimal cost, we employ the definition of offline optimum under PredSLS framework in Lemma 2, and obtain:

$$\begin{aligned}
J^{\star} &\leq \max(\|P\|_2, \|R\|_2) \sum_{t=0}^{T-1} \left(\|x_{t+1}^{\star}\|^2 + \|u_t^{\star}\|^2 \right) \\
&= \max(\|P\|_2, \|R\|_2) \sum_{t=0}^{T-1} \left(\left\| \sum_{k=0}^{t} \Phi_{t,k}^{x\mathrm{off}} w_{k-1} + \sum_{k=0}^{T} \widehat{\Phi}_{t,k}^{x\mathrm{off}} w_{k-1} \right\|^2 \right. \\
&\quad \left. + \left\| \sum_{k=0}^{t} \Phi_{t,k}^{\mathrm{uoff}} w_{k-1} + \sum_{k=0}^{T} \widehat{\Phi}_{t,k}^{\mathrm{uoff}} w_{k-1} \right\|^2 \right) \\
&\leq 2 \max(\|P\|_2, \|R\|_2) \sum_{t=0}^{T-1} 4 \sum_{k=0}^{T} \omega \phi \varrho^k \\
&= \frac{8 \max(\|P\|_2, \|R\|_2) \omega \phi}{1 - \varrho} \cdot T.
\end{aligned}$$

Hence $\upsilon J^{\star} = \upsilon O(T)$, now we can take $\upsilon = \Theta(\varrho^d)$ in (S.37) to complete the proof. □

APPENDIX D
EXPERIMENTAL DETAILS

A. Detailed performance comparison

Each run has $T = 180$ control steps. The first three columns of Fig. 1 use the same correlated disturbance process with reliable, homogeneously corrupted, or heterogeneously corrupted predictions. In the heterogeneous condition, the first three tracking agents or first twelve voltage buses receive reliable predictions, while the remaining sources have larger errors. The fourth column uses the separate time-varying-bias experiment.

Under heterogeneous errors, O-PredSLS reduces mean paired cost relative to Scalar by 34.5% in tracking and 14.8% in voltage regulation. Scalar performs better under homogeneous errors in both systems. These comparisons show where channel-dependent adaptation is useful: a common scalar must adjust reliable and unreliable predictions together. In the heterogeneous voltage condition, the cost of O-PredSLS nevertheless remains above Causal.

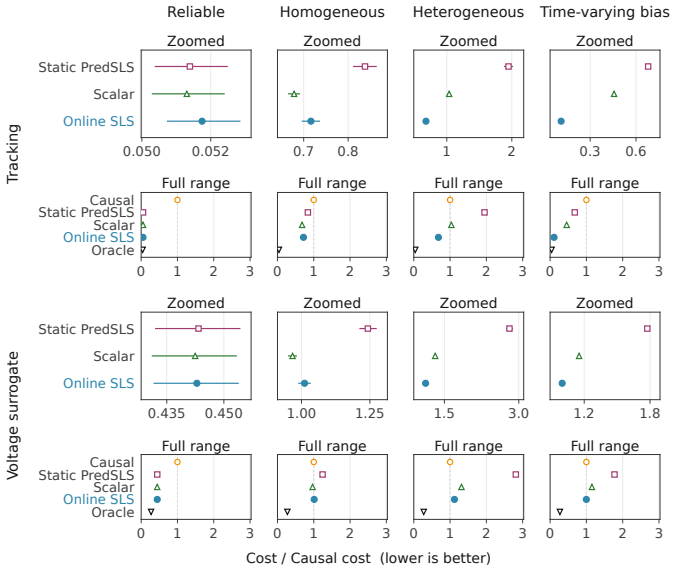


Fig. 1. Closed-loop cost under four synthetic prediction conditions. Points show mean per-run cost relative to Causal, with 95% intervals over 20 runs. The last uses reference motion, independent process noise, and a growing single-coordinate bias.

1) System models and offline design: The 5-agent tracking model is

$$A = 0.90I_{10} - 0.025(L_{\text{chain}} \otimes I_2),$$
$$B = I_{10}, \qquad Q = 7I_{10}, \qquad R = 0.35I_{10}, \tag{S.37}$$

where L_{chain} is the Laplacian of the 5-node path. For reference tracking, physical positions are $p_t = e_t + r_t$. Reference motion enters through (S.40), with no separate reference feedforward in the applied input.

The voltage model in Fig. 1 uses $A = I_{24}$ and

$$B = 2D_{\text{path}}\,\text{diag}(b_e)D_{\text{path}}^{\top} + 0.015I_{24},$$
$$b_e = 0.010 + 0.0025\,\text{depth}(\text{child}(e)), \tag{S.38}$$

where $[D_{\text{path}}]_{ie} = 1$ if edge e lies on the path from root 1 to bus i. The edges are

$$\begin{aligned}\mathcal{E} = \{&(1,2),(2,3),(3,4),(4,5),(4,6),(6,7),(4,8),\\ &(3,9),(9,10),(9,11),(11,12),(1,15),(15,13),\\ &(13,14),(13,16),(16,17),(16,18),(18,19),(19,20),\\ &(20,21),(18,22),(22,23),(23,24)\}.\end{aligned}$$

All buses are actuated. The model is a linear voltage regulation surrogate without inverter limits or hard voltage constraints. Voltage regulation uses $Q = 35I_{24}$ and $R = 0.08I_{24}$. For both models, P is the stabilizing DARE solution for (A, B, Q, R).

2) Prediction generation and information windows: The stationary conditions use independent latent processes and Gaussian prediction errors:

$$z_{t+1}^j = 0.72z_t^j + \sqrt{1-0.72^2}\,\eta_t^j,$$
$$w_t^j = a(0.35z_t^0 + 0.65z_t^j),$$
$$\widehat{w}_t^j = g_j w_t^j + as_j\epsilon_t^j, \tag{S.39}$$

where $z_0^j, \eta_t^j, \epsilon_t^j$ are independent standard normal variables, $a = 0.35$ for tracking, and $a = 0.006$ for voltage. Reliable and homogeneous conditions use $(g_j, s_j) = (1, 0.05)$ and $(0.65, 0.65)$, respectively. The heterogeneous condition uses $(1, 0.05)$ for the reliable sources specified in the main text and $(0.15, 1.60)$ elsewhere. Initial states are $x_0^i = (-1)^{i-1}(0.12, -0.08)^{\top}$ for tracking and zero for voltage.

The time-varying-bias condition uses separate reference-tracking runs:

$$w_t = Ar_t - r_{t+1} + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, \sigma^2 I),$$
$$\widehat{w}_t = w_t + b\frac{t}{T}\big(4 + \sin(0.4t)\big)\mathbf{e}_1. \tag{S.40}$$

Tracking uses $(\sigma, b) = (0.75, 1)$. Voltage uses $(\sigma, b) = (0.75a_v, a_v)$, where $a_v = 0.006$, zero initial error, and

$$r_t^j = 1 + 0.01\sin\big(2\pi t/T + 2\pi(j-1)/24\big). \tag{S.41}$$

Predictions are generated once and remain fixed. The available indices are $\max\{0, t-d\} \le q < \min\{T, t+d-1\}$. The archived realization associates disturbance index q with predictive column $k = q + 1$ and stores only $1 \le k < T$.

3) Predictive-response updates and parameter selection: At action time $t \ge 30$, the update uses the revealed pairs $(\widehat{w}_s, w_s)$ for $s = \underline{t}, \dots, t-1$. Let $\lambda_{\text{e}} > 0$ denote the experimental ridge parameter. For each available column k and $a \in \{x, u\}$, Scalar uses the corresponding update

$$\widehat{\Phi}_{t,k}^{a}(t) = \frac{\sum_{s=\underline{t}}^{t-1} w_s^{\top}\left(\Phi_{t,s}^{a,\text{off}\top}\Phi_{t,s}^{a,\text{off}}\right)\widehat{w}_s + \lambda_{\text{e}}\|\Phi_{t,s}^{a,\text{off}}\|_F^2}{\sum_{s=\underline{t}}^{t-1}\|\Phi_{t,s}^{a,\text{off}}\widehat{w}_s\|_2^2 + \lambda_{\text{e}}\|\Phi_{t,s}^{a,\text{off}}\|_F^2}\widehat{\Phi}_{t,k}^{a,\text{off}}.$$

Before $t = 30$, both methods use the offline predictive coefficients. The fixed ridge parameters are $\lambda_{\text{e}} \simeq 2.45 \times 10^{-2}$ for tracking and $\lambda_{\text{e}} \simeq 1.27 \times 10^{-5}$ for voltage.

4) Communication selection from predictive responses: The communication experiment retains the voltage input matrix in (S.38) and the evaluation weights, but uses

$$A = 0.4I + 0.58D^{-1/2}WD^{-1/2},$$
$$D = \text{diag}(W\mathbf{1}),$$
$$W_{ij} = \begin{cases}\exp(-d_{ij}/2), & i \ne j,\\ 0, & i = j,\end{cases}$$

where d_{ij} is the tree distance and $\rho(A) = 0.98$. For the reference in (S.41), set

$$w_t = A(r_t - \mathbf{1}) - (r_{t+1} - \mathbf{1}) + \xi_t,$$
$$\widehat{w}_t^j = w_t^j + \beta\sigma a_j\epsilon_t^j,$$

where r_t is given by (S.41), $\xi_t \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 0.75(0.006/0.35)$, and the ϵ_t^j are independent standard normal variables, independent of ξ_t. We set $a_j = 4$ for $j \in \{3, 6, \dots, 24\}$ and $a_j = 0.05$ otherwise.

From the 6 discovery runs, the heatmaps report

$$E_{qj} = \frac{1}{\sigma}\left[\frac{1}{6}\sum_{s=1}^{6}(\widehat{w}_q^{s,j} - w_q^{s,j})^2\right]^{\frac{1}{2}},$$
$$S_{ij} = \frac{1}{6}\sum_{s=1}^{6}\sum_{t=30}^{T-1}\sum_{q=0}^{T-1}\|[\widehat{\Phi}_{t,q}^{(s)}]_{ij}\|_F^2.$$

Their display is saturated at 500, as shown in Figure 6 in the main body.

REFERENCES

[1] Y. Wu, J. Yu, and T. Li, "System-level performance and communication tradeoff in networked control with predictions," *arXiv preprint arXiv:2508.13475*, 2025.